frontiers | Frontiers in Sustainability

TYPE Original Research
PUBLISHED 26 November 2025
DOI 10.3389/frsus.2025.1681014

Check for updates

OPEN ACCESS

EDITED BY
Farshid Pahlevani,
University of New South Wales, Australia

REVIEWED BY
Luisa Fernanda Duque-Buitargo,
University of the Valley, Colombia
Ariovaldo Silva,
State University of Campinas, Brazil

*CORRESPONDENCE
Joseph G. Usack
joseph.usack@uga.edu

RECEIVED 06 August 2025
REVISED 10 September 2025
ACCEPTED 11 November 2025
PUBLISHED 26 November 2025

CITATION
Taiwo KJ, Ogundipe SO, Kerr WL, Pegg RB, Suh JH and Usack JG (2025) Sustainable management of simulated coffee processing wastewater using micro-aeration enhanced anaerobic digestion: a long-term technical evaluation.
Front. Sustain. 6:1681014.
doi: 10.3389/frsus.2025.1681014

COPYRIGHT
© 2025 Taiwo, Ogundipe, Kerr, Pegg, Suh and Usack. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

Sustainable management of simulated coffee processing wastewater using micro-aeration enhanced anaerobic digestion: a long-term technical evaluation

Kayode J. Taiwo[1], Samuel O. Ogundipe[1], William L. Kerr[1], Ronald B. Pegg[1], Joon Hyuk Suh[1] and Joseph G. Usack[1,2,3]*

[1]Department of Food Science and Technology, University of Georgia, Athens, GA, United States, [2]New Materials Institute, University of Georgia, Athens, GA, United States, [3]Institute for Integrative Precision Agriculture, Office of Research, University of Georgia, Athens, GA, United States

Introduction: Coffee bean production generates high volumes of contaminated processing water in regions of the world that often lack the necessary infrastructure to provide adequate treatment. This coffee processing water (CPW) contains high organic loads alongside ecopharmokinetic and recalcitrant compounds, such as caffeine and tannins, which, when discharged, pollute the environment and degrade freshwater supplies that nearby populations may depend on.

Methods: In this study, an anaerobic digestion (AD) reactor and a micro-aeration enhanced anaerobic digestion (MA-AD) reactor were operated in parallel for 430 days to compare their effectiveness in treating and valorizing simulated CPW to promote a more sustainable approach to coffee production. In coffee-producing regions where access to centralized wastewater treatment facilities is limited, MA-AD offers a technologically and economically accessible option that can be implemented by industrial-scale coffee processors as well as small- to medium-scale processors in rural settings with limited technical infrastructure. To test this, oxygen was intermittently dosed into an AD microbiome, allowing for a comparative assessment across anaerobic and micro-aerobic redox regimes.

Results: The conventional AD and MA-AD achieved comparable reductions in total and volatile solids (>48 and >60%, respectively) and total and soluble chemical oxygen demand (>66 and >86%, respectively), MA-AD exhibited significantly higher total suspended solids concentrations and turbidity in later phases, likely due to gas sparging-induced floc disruption and particulate release. pH profiles indicated a shift towards increased acidification under MA-AD, without compromising process stability, with both reactors stabilizing between pH 6.8 and 7.1. Caffeine degradation was accelerated under MA-AD at the lowest O_2 dosing level (28.73 ± 1.10 mL $O_2 \cdot d_{-1} \cdot L_{reactor-1}$) after spiking caffeine (>85% removal in 28 h), and decreased at higher O_2 dosing levels (100.73 ± 6.90 mL $O_2 \cdot d_{-1} \cdot L_{reactor-1}$; 228.60 ± 3.92 mL $O_2 \cdot d_{-1} \cdot L_{reactor-1}$). Finally, methane production was consistently lower in MA-AD, attributed to the oxygen sensitivity of methanogens and possible substrate competition.

Conclusion: These results underscore the importance of oxygen dose regulation, redox control, and microbial adaptation in optimizing MA-AD performance. The findings support MA-AD as a promising strategy for ensuring a safer and more

sustainable water supply by enhancing the treatment of ecopharmokinetic and recalcitrant compounds in CPW.

KEYWORDS

sustainable coffee production, clean water, responsible production, effluent quality, microaeration, caffeine degradation

1 Introduction

Coffee is one of the most globally traded commodities, with trade volumes expected to exceed 10.5 million tons in 2025 (United States Department of Agriculture-Foreign Agricultural Service, 2024). However, the negative externalities of coffee production are disproportionately encumbered by citizens of coffee-producing countries, such as those in Africa, the Americas, and Asia, where most of the world's coffee is grown and processed (Utrilla-Catalan et al., 2022). These are often remote and agrarian regions that lack the necessary infrastructure, resources, and regulatory support to properly treat the waste streams generated during coffee bean processing, resulting in widespread environmental pollution and public health risks (Dadi et al., 2018). Indeed, sustainable and fair coffee production has become a critical issue that has attracted global attention. More and more consumers are demanding that major coffee retailers, such as Nestle, JDE Peets, and Starbucks, source their coffee in a more environmentally and socially responsible way.

A significant issue undermining sustainable coffee production is its high water footprint. Studies estimate coffee bean production consumes 18,400 $L{\cdot}kg^{-1}$ of fresh water before reaching the roasting stage, accounting for 68% of the total water consumption in the life cycle of ground coffee (Ratchawat et al., 2020). The wet processing method of coffee beans uses copious volumes of water to remove pulp and wash the coffee cherries after fermentation, generating a dark, turbid, low-pH wastewater high in organic matter, which is unsuitable for crop irrigation. Alongside the biodegradable fraction, the coffee processing water (CPW) also contains ecopharmokinetic and biologically recalcitrant compounds, such as alkaloids (e.g., caffeine, theobromine, trigonelline), chlorogenic acids, and tannins (Ijanu et al., 2020). Caffeine and its chlorinated derivatives have been shown to adversely affect the central nervous and digestive systems of animals (Mazzafera, 2002). In terrestrial ecosystems, its presence in soil can suppress seed germination and hinder early plant development, posing a risk to soil fertility and crop establishment (Chen et al., 2018; Nanjundaiah et al., 2017). Its widespread occurrence has led to its classification as an emerging organic contaminant (Chen et al., 2018; Rodriguez-Narvaez et al., 2017). When discharged into the environment, CPW finds its way into local water bodies, where it depletes dissolved oxygen (DO) via eutrophication, killing aquatic biota, while stimulating the proliferation of harmful microbial populations (Figueroa Campos, 2022). Indeed, studies from coffee-producing areas consistently report elevated total suspended solids (TSS) and chemical oxygen demand (COD) concentrations in nearby water bodies (2,260 $mg{\cdot}L^{-1}$ and 50,000 $mg{\cdot}L^{-1}$, respectively) (Dadi et al., 2018; Rattan et al., 2015). In addition to causing ecosystem damage, CPW-contamination poses a human health crisis in the rural agrarian communities whose people rely on this freshwater for daily life (Figueroa Campos, 2022; Haddis and Devi, 2008). Studies report that villagers who use these affected water bodies for domestic purposes experience spinning sensations, eye/skin irritation, breathing problems, and nausea (Haddis and Devi, 2008), most likely due to the organic acids and residual caffeine in CPW. Needless to say, current CPW management practices do not comport with the goals of sustainable development. Innovative solutions and policies are needed to promote responsible coffee production and ensure equitable access to clean water in these regions.

Anaerobic digestion (AD) is a well-established biological treatment technology that is already widely applied in rural and resource-constrained regions, due to its low cost and technical simplicity (Alemayehu et al., 2020). The key advantage of AD is its ability to simultaneously reduce the organic load of waste streams and generate renewable energy in the form of methane-rich biogas (Angenent et al., 2022; Zhou et al., 2024). The biogas resulting from CPW treatment could fuel a combined heat and power (CHP) generator, providing heat and electricity to remote coffee bean processors that may lack access to conventional or reliable energy sources (Usack et al., 2014). Coffee bean processing requires heat for coffee bean drying and roasting, as well as electricity for dehulling and grinding. Altogether, energy consumption constitutes up to 30% of coffee production's carbon footprint (Ratchawat et al., 2020). Therefore, implementing AD to treat CPW during coffee bean processing represents an opportunity to promote responsible water resource management while decarbonizing the coffee supply chain. However, despite these advantages, AD's efficacy in treating ecopharmokinetic and recalcitrant compounds is limited (Gomes de Barros et al., 2020), resulting in low COD removal, accumulation of volatile fatty acids (VFAs), and even system failure under mono-digestion conditions (Qiao et al., 2015). While co-digestion and nutrient supplementation have been explored to improve AD performance (Du et al., 2020; Selvamurugan et al., 2010), these approaches are ineffective in degrading problematic compounds to safe levels.

A potential solution to this problem is a slightly more advanced AD process, known as micro-aeration. Micro-aeration involves introducing trace amounts of air into the anaerobic digester to promote the growth of facultative bacteria. Unlike strict anaerobes, these facultative bacteria produce potent enzymes that more effectively degrade biologically recalcitrant compounds (Nguyen and Khanal, 2018). Indeed, multiple studies have shown that micro-aeration enhances the degradation of recalcitrant compounds similar to those found in CPW (Díaz et al., 2011; Fu et al., 2023;

Abbreviation: AD, Anaerobic digestion; BOD, Biochemical oxygen demand; CPW, Coffee processing wastewater; COD, Chemical oxygen demand; DO, Dissolved oxygen; NTU, Nephelometric turbidity units; MA-AD, Micro-aerated anaerobic digestion; ORP, Oxidation–reduction potential; TS, Total solids; TSS, Total suspended solids; TVSS, Total volatile suspended solids; VFAs, Volatile fatty acids; VS, Volatile solids.

Magdalena et al., 2022). In addition to augmenting the removal of specific compounds, micro-aeration has been reported to improve the overall performance and stability of the AD process by increasing biogas yields, COD removal, attenuating VFAs, and reducing total sludge volumes, among other benefits (Ding et al., 2024; Fu et al., 2023; Xu et al., 2021). Finally, micro-aeration represents a cost-effective strategy that can be readily implemented in existing or future AD operations in remote coffee-producing regions. However, despite this apparent opportunity to improve the sustainability management of coffee production, no research has evaluated the effectiveness of using micro-aeration to treat coffee processing residues. Furthermore, long-term stability is crucial to the adoption of any micro-aeration-enhanced anaerobic digestion (MA-AD) process; however, many bench studies are short in duration, limiting their scope of insight regarding the robustness, reliability, and practicality of the technology in the context of real-world applications.

Therefore, this study aims to evaluate the performance of MA-AD in treating CPW. Specifically, the study involves operating two parallel reactors continuously for 430 days, spanning multiple operating phases, to optimize oxygen dosage and assess the impact of micro-aeration on methane yield, COD, solids, and turbidity reduction, as well as the degradation of ecopharmokinetic and recalcitrant compounds. By providing insights into the underlying mechanisms and operational conditions, this research seeks to contribute to the broader goal of sustainable water resource management in the coffee industry.

2 Materials and methods

2.1 Reactor set-up

The reactors consisted of two benchtop fermenters (BioFlow3100, New Brunswick Scientific Co., Edison, NJ, United States), each with a 1.5-L vessel and an operational volume of 1 L. Each reactor was outfitted with a headplate that included a sampling port, a gas sparger, an oxygen-reduction potential (ORP) probe (Sensorex, Garden Grove, CA, United States), an influent inlet, and an effluent outlet. The reactors were set up as described by Usack et al. (2012) and operated as continuous stirred tank reactors (CSTR) using an overhead stirrer (Fisherbrand Compact Digital, Waltham, MA, United States) integrated with an ORP controller (Oakton 220, Vernon Hills, IL, United States). A 3-L water bath (Fisher Scientific, Hampton, NH, United States) maintained the reactor temperature at 37 ± 1 °C. The effluent was withdrawn using a peristaltic pump (Masterflex®, Vernon Hills, IL, United States) for further analysis. Biogas production was monitored with integrated flow meters (BPC Instruments, Mobilvägen, LU, Sweden). The reactors comprised (1) a control reactor system for conventional AD and (2) an experimental reactor system for MA-AD (Figures 1a,b). The reactors were allowed to reach pseudo-steady-state operation before initiating the experimental phase of the study. During the experimental phase, one reactor was sparged with oxygen to induce microaerobic conditions, while the other was supplied with 100% nitrogen gas (N_2) to maintain anaerobic conditions. The ORP in the micro-aeration reactor was regulated using integrated ORP controllers following the guidelines of Nguyen and Khanal (2018).

2.2 ORP-controlled micro-aeration system

In this study, the experimental reactor was controlled by an ORP controller to maintain micro-aerobic conditions by targeting ORP increments +20 mV higher than the baseline anaerobic ORP during each study phase (i.e., Phase 1 = +20 mV; Phase 2 = +40 mV; Phase 3 = +60 mV). Micro-aeration was achieved by injecting oxygen at a controlled flow rate of 10 mL·min^{-1} using a peristaltic pump. The daily oxygen dosing volume was tracked using integrated flow meters, which measured the oxygen flow before it entered the reactor, ensuring precise and accurate oxygen measurements. The ORP probe was highly sensitive to DO, having a detection limit as low as 0.1 mg O_2·L^{-1}. As oxygen was injected, the ORP increased, indicating a shift towards a more oxidative environment. The ORP-controlled micro-aeration system was fully automated, with oxygen injection cycles triggered when the ORP fell below the target value and ceasing once the target ORP was achieved. This system maintained stable micro-aerobic conditions in the reactor, with consistent fluctuations of ORP around the target set point. The ORP setpoint values and the average oxygen dosing amount during each experimental stage are presented in Table 1.

2.3 Inoculum and CSTR operation

The two reactors were inoculated with waste-activated sludge (WAS) sourced from a wastewater treatment plant (Gwinnett, GA, United States). Coffee bean residues (CBR) were obtained from a dining hall at the University of Georgia, Athens, United States, and steeped for 48 h at room temperature before being used as a model CPW substrate. Reconstituted whole milk powder (RWM) was used in place of milk processing wastewater as a co-digestion substrate to provide additional organic content, buffering capacity, and nutrients. The final substrate was a mixture containing per liter: CBR, 700 mL; WAS, 285.6 mL; RWM, 6.9 g; yeast, 1 g; 7.2 mL of mineral stock; and 7.2 mL of trace element solution. The mineral stock contains the following (in g·L^{-1}): $FeCl_2·4H_2O$, 370; $MgCl_2·6H_2O$, 120; KCl, 86.7; NH_4Cl, 26.6; and $CaCl_2·2H_2O$, 16.7. The trace element solution consists of (in g·L^{-1}): $COCl_2·6H_2O$; $MnCl_2·4H_2O$, 1.33; H_3BO_3, 0.38; $ZnSO_4·7H_2O$, 0.29; $Na_2MoO_4·2H_2O$, 0.17; and $CuCl_2·2H_2O$, 0.18. The reactors were maintained at a constant temperature of 37 ± 1 °C and operated with a hydraulic retention time of 40 days at an OLR of 0.6 g COD·L^{-1}·d^{-1}. The influent composition comprised 20% CBR, 60% RWM, and 20% WAS (i.e., COD basis). The substrate compositions are shown in Table 2.

2.4 Analyses

Biogas production volume and effluent pH were measured daily. Total solids (TS), volatile solids (VS), total COD, and soluble COD (0.45-µm filtrate) were measured weekly in triplicate (i.e., technical replicates) and reported as mean ± SD. Total dissolved solids (TDS), total dissolved volatile solids (TDVS), total suspended solids (TSS), and total volatile suspended solids (TVSS) were measured at the end of each treatment phase, also in triplicate. All analyses were performed according to standard methods (American Public Health Association, 2023). Headspace gas samples were taken weekly to assess biogas composition

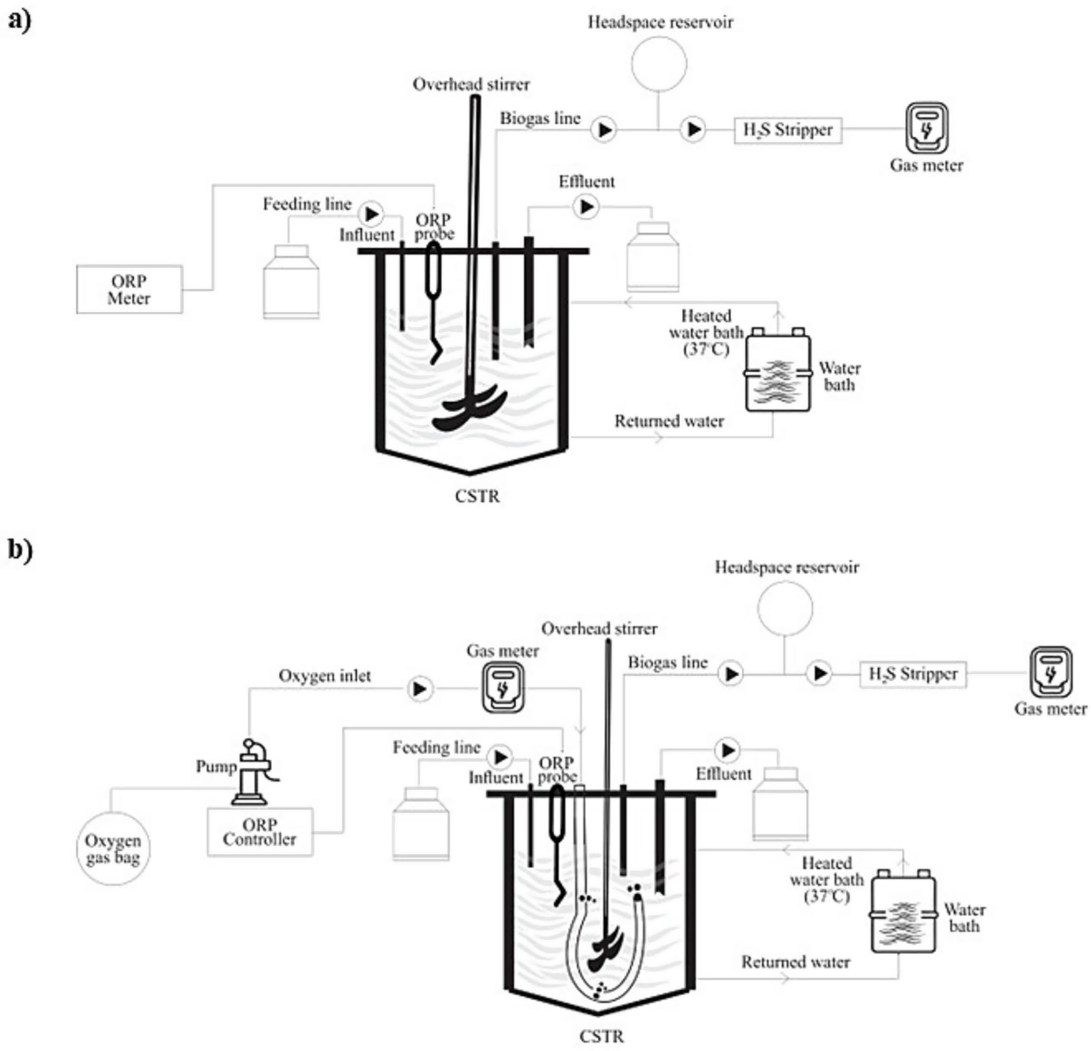


FIGURE 1
Schematic representation of experimental setups used in the study. **(a)** Conventional anaerobic digestion (AD) reactor equipped with an oxidation–reduction potential (ORP) meter for monitoring redox conditions. **(b)** Micro-aerated anaerobic digestion (MA-AD) reactor modified with a custom-designed ORP controller and automated oxygen injection system to regulate micro-aeration. Both systems utilize continuously stirred tank reactors (CSTRs), integrated with biogas lines, hydrogen sulfide (H_2S) strippers, overhead stirring mechanisms, and temperature-controlled water baths (37 °C) to maintain mesophilic conditions. CSTR, continuously stirred tank reactor; H_2S, hydrogen sulfide.

(i.e., CH_4, CO_2), using a gas chromatograph equipped with a methanizer and a flame ionization detector (8610C, SRI Instruments, Torrance, CA, United States). Nitrogen served as the carrier gas, with an inlet and detector temperature maintained at 110 °C and a constant oven temperature set at 40 °C. The concentrations of individual VFAs were determined weekly using gas chromatography with a flame ionization detector (Agilent 6890, Santa Clara, CA, United States), following the procedure described by Usack et al. (2014) with minor modifications. Hydrogen was used as the carrier gas, with the inlet temperature set to 200 °C and the detector temperature set to 275 °C. Individual VFA species were separated using a fused silica capillary column (NUKOL, 15 m × 0.53 mm × 0.50 µm film thickness; Supelco Inc., Bellefonte, PA, United States). The initial temperature was maintained at 70 °C for 2 min, followed by an increase of 12 °C/min until reaching 200 °C, at which point it was held for an additional 2 min.

Turbidity was measured in triplicate at the end of each treatment phase using a turbidity meter (HI 98703, Hanna Instruments, Woonsocket, RI, United States), following the manufacturer's instructions. Chromophoric compounds were also quantified using UV–Vis spectrophotometry (Cary 60, Agilent Technologies, Santa Clara, CA, United States) equipped with a diode array detector. Raw CPW was scanned from 200 to 400 nm, revealing a maximum absorbance at 270 nm. A calibration curve was established from serial dilutions of CPW, and effluent samples were analyzed accordingly. Caffeine was extracted using a liquid–liquid extraction method (Vandeponseele et al., 2021) in triplicate for each treatment phase and for multiple time points during the caffeine spiking trials. Caffeine degradation rates were evaluated in three independent spike tests at the end of each treatment phase. Analytical-grade caffeine (≥99%, Sigma-Aldrich, St. Louis, MO, United States) was dissolved in deionized water and dosed to each reactor to a target concentration of ~200 $mg{\cdot}L^{-1}$. Caffeine was quantified using an external calibration curve over the range of 1–2,000 $mg{\cdot}L^{-1}$. The regression equation was $y = 46.538x - 1.9126$

TABLE 1 Operational parameters of micro-aeration applied during each experimental phase in the micro-aerated anaerobic digestion (MA-AD) reactor.

Phase	Set ORP (mV)	Dosing techniques	Average oxygen dosing (mL $O_2 \cdot d^{-1} \cdot L_{reactor}^{-1}$)	Gas type	Duration (days)
Zeroth	−490	N/A	NA	N/A	160 (Days 0–159)
First	−470	Automatic, on-demand	28.70 ± 1.03	Air	129 (Days 160–289)
Second	−450	Automatic, on-demand	100.70 ± 6.90	Oxygen	70 (Days 290–360)
Third	−430	Automatic, on-demand	228.60 ± 3.92	Oxygen	70 (Days 361–430)

The table summarizes the oxidation–reduction potential (ORP) setpoints, oxygen dosing strategies, average oxygen volumes delivered per liter of reactor volume per day, gas type used, and duration of each phase. The zero phase served as a baseline without aeration, followed by controlled micro-aeration using either air or pure oxygen.

TABLE 2 Physicochemical characterization of substrates used in the study: coffee bean residue (CBR), waste activated sludge (WAS), and reconstituted whole milk powder (RWM).

Parameters	Units	CBR	WAS	RWM
COD	$g \cdot L^{-1}$	4.00 ± 1.09	8.42 ± 0.59	1.64 ± 2.41
TS	%	0.34 ± 0.05	3.03 ± 0.10	97.44 ± 0.01
VS	%	0.22 ± 0.04	2.30 ± 0.65	86.31 ± 0.22
pH	-	5.70 ± 0.20	6.20 ± 0.42	6.70 ± 0.12
TDS	%	0.13 ± 0.01	0.34 ± 0.01	N/A
TVDS	%	0.10 ± 0.00	0.16 ± 0.00	N/A
TSS	%	0.17 ± 0.00	2.10 ± 0.05	N/A
TVSS	%	0.14 ± 0.03	1.40 ± 0.03	N/A

Parameters include chemical oxygen demand (COD), total solids (TS), volatile solids (VS), pH, total dissolved solids (TDS), total volatile dissolved solids (TVDS), total suspended solids (TSS), and total volatile suspended solids (TVSS). Data are presented as mean ± standard deviation.

with an R^2 of 0.9999, indicating excellent linearity. The limit of detection and limit of quantification were 13.86 $\mu g \cdot L^{-1}$ and 42.01 $\mu g \cdot L^{-1}$, respectively. The supernatants from centrifuged samples underwent three chloroform/methanol extractions, followed by solvent removal using a Büchi R-210 Rotavapor connected to a V-850 vacuum controller and a V-700 vacuum pump. The residue was resolubilized in HPLC-grade water, filtered (0.45-µm nylon), and analyzed via HPLC (Model 1100, Agilent Technologies, Santa Clara, CA, United States) with a variable wavelength detector. Separation was achieved using a reversed-phase Luna C18(2) column (4.6 × 250 mm, 5 µm particle size; Phenomenex, Torrance, CA, United States). An isocratic mobile phase was employed comprising water/methanol (65/35, v/v) with 1% acetic acid, a 1 $mL \cdot min^{-1}$ flow rate, and a 20 µL injection volume. Caffeine was detected at 280 nm and quantified via calibration curves.

2.5 Statistical analysis

Statistical differences between the AD and MA-AD reactors were evaluated using one-way analysis of variance (ANOVA), followed by Tukey's Honest Significant Difference (Tukey's HSD) post-hoc test for pairwise comparisons. All analyses were conducted in R software (version 4.3.1). The stats package was used for ANOVA, and the agricolae and multicomp packages were employed for post-hoc tests and multiple comparisons. Data sets for methane yield, total solids (TS), VS, COD, and pH were assessed for normality and homogeneity of variance using the Shapiro–Wilk and Levene's tests, respectively. Where assumptions were not met, appropriate transformations or non-parametric equivalents were applied. All statistical tests were performed at a significance level of $\alpha = 0.05$. Data are presented as mean ± standard deviation unless otherwise stated.

3 Results and discussion

3.1 Micro-aeration alters pH dynamics without compromising reactor stability or organic matter removal rates during coffee processing water treatment

3.1.1 pH fluctuations in MA-AD were linked to changes in CO_2 equilibria and microbiome responses

The pH levels in the AD and MA-AD reactors were monitored daily throughout the study to assess the impact of micro-aeration on reactor stability. During the start-up phase (Phase 0), both reactors stabilized at a baseline pH of 7.08 ± 0.02 (Figure 2), indicating that the AD process was functioning normally and consistently. During the stabilization phase, the main performance parameters (i.e., pH, biogas production, VFA concentration) varied less than ±10% of the running average, indicating consistent pseudo-steady-state operation between reactors (Usack et al., 2012). Initially, air was introduced as the dosing gas to facilitate micro-aeration in the MA-AD reactor. However, between Days 160 and 290, a reduction in CO_2 partial pressure in the reactor headspace due to CO_2 stripping and headspace dilution led to a gradual increase in pH. This pH shift was most likely caused by chemical changes related to gas–liquid CO_2 equilibria rather than biological processes, as the rise in pH was not accompanied by any significant alterations in VFA concentration or methane production. Air is ~78% N_2 and 21% O_2; therefore, dosing air rather than pure O_2 causes significantly more CO_2 stripping and headspace dilution, requiring ~5 times the volumetric flow rate to achieve the same oxygen dosing rate. Pure oxygen gas was introduced on Day 280 to address the pH imbalance, replacing air. Following this intervention, a greater decrease in pH was observed in the MA-AD reactor compared to the AD reactor. This drop below baseline levels can be attributed to enhanced microbial activity under microaerobic conditions, promoting the production of organic acids by facultative microorganisms, as reported by Lim and Wang (2013). The switch from air to pure oxygen likely created more favorable conditions for the proliferation of facultative bacteria (Lim and Wang, 2013), thereby intensifying acid production and maintaining higher levels of dissolved CO_2.

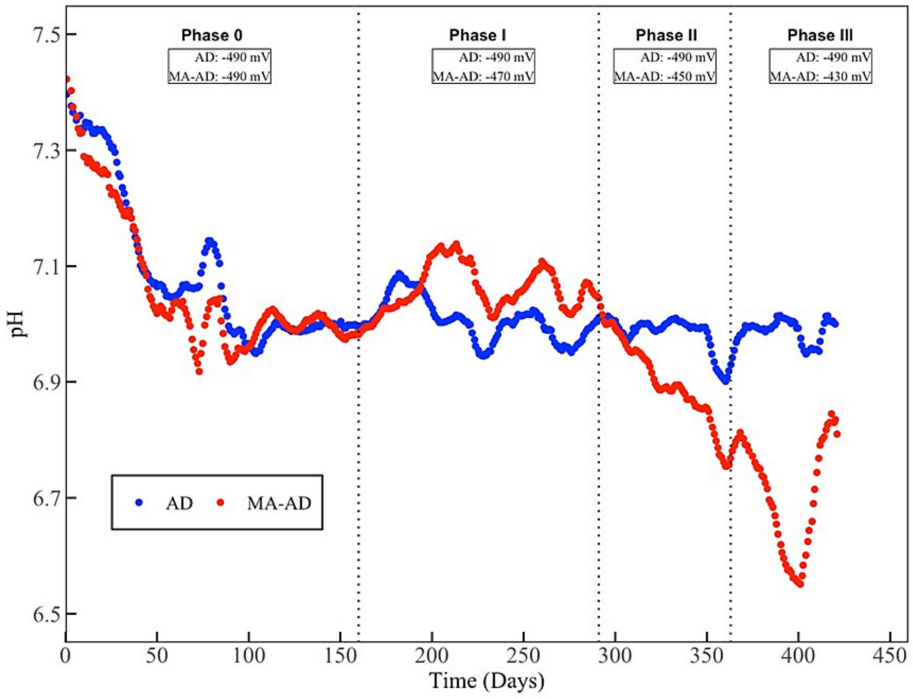


FIGURE 2
Temporal variation of pH in conventional anaerobic digestion (AD) and micro-aerated anaerobic digestion (MA-AD) reactors over the 430-day operational period. The experiment was divided into four phases (Phases 0–III), each characterized by distinct oxidation–reduction potential (ORP) setpoints for AD and MA-AD reactors. The MA-AD system exhibited more pronounced pH fluctuations, particularly in later phases, reflecting the dynamic response to micro-aeration. AD, anaerobic digestion; MA-AD, micro-aerated anaerobic digestion.

Throughout the study, caffeine was periodically introduced into the reactors to elevate its concentration, which, in turn, influenced the pH dynamics. On Day 230 (Phase I), both reactors were spiked with caffeine at a concentration of 200 mg·L^{-1} to achieve higher substrate levels and assess the effects of micro-aeration on caffeine degradation, as the steady-state caffeine concentrations in the reactor broths were low (i.e., 1.3 mg·L^{-1}). This spiking event led to a marked drop in pH between Day 230 and Day 249, which could be attributable to (1) the inhibition of methanogenesis or (2) the microbial breakdown of caffeine into acidic metabolites such as theobromine, theophylline, paraxanthine, and various low-molecular-weight organic acids. The transient accumulation of these degradation products increased system acidity, resulting in a pH decline. A second caffeine spike of 200 mg·L^{-1} was administered between Day 352 and Day 369 (Phase II) to further assess the influence of micro-aeration on caffeine degradation. Consistent with the first spike, this intervention induced another drop in pH. Subsequently, between Day 403 and Day 411, a third caffeine spiking event led to an additional pH decrease in both reactors. Despite these drops, the reactors stabilized after a few days, particularly in the MA-AD reactor. From Day 395 onwards, the MA-AD reactor re-stabilized at 6.8, following an initial drop to 6.56 (Figure 2).

The consistent recovery of reactor pH after repeated caffeine spikes, particularly in the MA-AD reactor, shows the system's resilience under chemical and operational stress. This inherent stability is crucial for deployment in decentralized or low-infrastructure settings, where automated control systems may be limited. From a sustainability standpoint, the ability to self-stabilize without corrective chemical inputs contributes to reduced operational costs and environmental impacts.

3.1.2 Stable organic matter removal suggests efficient substrate turnover under micro-aerobic conditions

The COD removal efficiencies were evaluated to assess the degradation of organic matter under anaerobic and micro-aerated conditions. Throughout the experimental phases, the total COD (TCOD) removal efficiency was 61.32 ± 17.98% for the AD reactor and 63.90 ± 17.29% for the MA-AD reactor (Figure 3). Similarly, soluble COD (SCOD) removal efficiencies were 88.29 ± 11.29% and 91.04 ± 7.04% for the AD and MA-AD reactors, respectively. Statistical analysis revealed no significant differences between the two reactors in TCOD removal ($p = 0.58$) and SCOD removal ($p = 0.24$). Furthermore, when COD removal efficiencies were analyzed for the experimental phases alone, the differences remained statistically insignificant ($p = 0.413$ for TCOD and $p = 0.719$ for SCOD). These findings suggest that the shift in acid production pathways, if any, due to micro-aeration, neither enhanced nor impaired the overall degradation of organic matter. The results align with the findings of Duarte et al. (2024), who similarly reported no significant differences in SCOD removal between micro-aerated and non-micro-aerated reactors.

An increase in hydrolysis rates typically causes an increase in the steady-state SCOD concentration. However, the MA-AD reactor exhibited a 2.85% lower average SCOD concentration than the AD reactor. This modest SCOD reduction, combined with lower methane

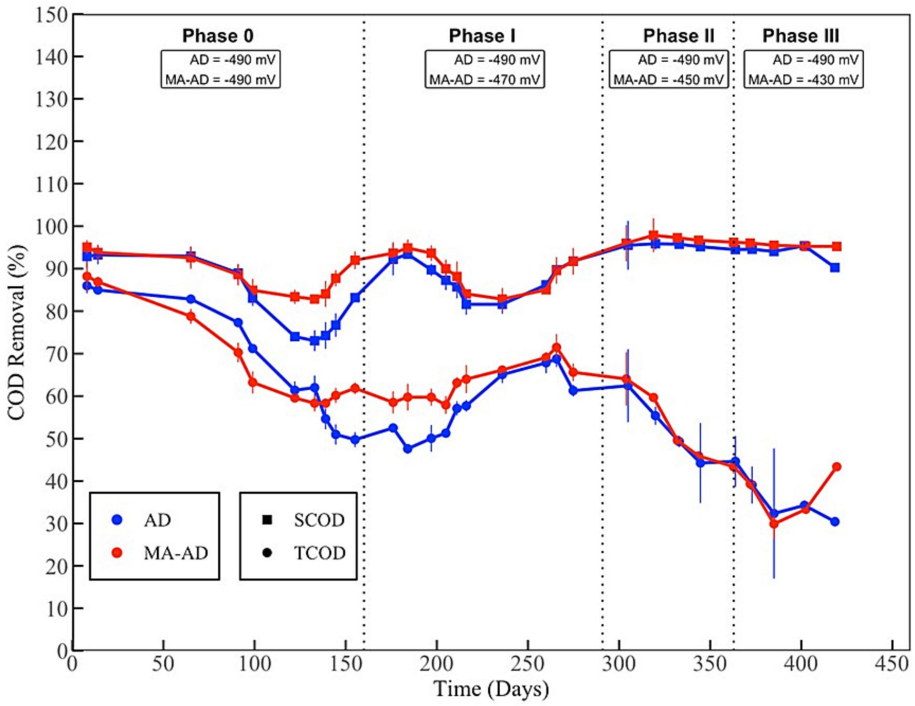


FIGURE 3
Removal efficiency of chemical oxygen demand (COD) over time in conventional anaerobic digestion (AD) and micro-aerated anaerobic digestion (MA-AD) reactors. Both soluble COD (SCOD) and total COD (TCOD) were monitored across four operational phases, each defined by specific oxidation–reduction potential (ORP) setpoints. Both systems exhibited similar overall COD removal trends, with no statistically significant differences observed. Error bars represent the standard deviation of technical replicates. SCOD, soluble chemical oxygen demand; TCOD, total chemical oxygen demand.

yields, suggests micro-aeration had (1) no significant effect on hydrolysis rates or (2) increased hydrolysis rates, but the hydrolysis products were quickly metabolized (Harirchi et al., 2022), being assimilated into microbial biomass (Fernández-Domínguez et al., 2023) or fully oxidized to non-methane end products (i.e., CO_2, H_2O) (Gaballah et al., 2025; Harirchi et al., 2022). Hence, the absence of elevated SCOD does not preclude an increase in hydrolytic activity under micro-aerobic conditions but could imply more efficient substrate turnover or a diversion of intermediates away from methane production.

3.1.3 No significant VFA build-up indicates a balanced trophic cascade during micro-aeration

The VFA concentrations were closely monitored to evaluate the potential impacts of micro-aeration on acidogenesis and methanogenesis. VFA levels remained stable throughout the study, with concentrations of 364.80 ± 49.40 mg·L^{-1} in the MA-AD reactor and 383.90 ± 88.20 mg·L^{-1} in the AD reactor. Despite notable pH shifts caused by changes in gas dosing strategies, no significant VFA accumulation was observed in either reactor, indicating a stable microbiome. Under micro-aerobic conditions, the system likely favored the production of non-VFA organic acids or other acidic intermediates, such as lactic acid, succinic acid, or similar compounds, which were not directly measured. If micro-aeration stimulated VFA generation through increased acidogenic activity, it must have simultaneously increased VFA catabolism, as no VFA accumulation occurred. These findings are consistent with those reported by Canul Bacab et al. (2020), who observed comparable shifts in microbial metabolic pathways under oxygen-limited environments. Additionally, the low OLR used in this study likely contributed to maintaining reactor stability by limiting acetate availability for methanogenesis, which prevented VFA accumulation. This observation supports the findings of Fu et al. (2023), who reported that the influence of micro-aeration on VFA accumulation is closely linked to oxygen dosing rates and OLR.

These findings reinforce the suitability of MA-AD as a sustainability-enhancing technology for agro-industrial wastewater treatment. Its ability to handle high-strength effluents, while maintaining reactor stability and moderate COD removal, positions it as a modular solution adaptable to both industrial-scale facilities and smallholder operations. In addition, the need for advanced controls, chemical additives, or centralized infrastructure is minimal, which further positions MA-AD as a promising approach to reducing environmental contamination, improving water safety, and providing broader access to clean processing technologies.

3.2 Micro-aerobic conditions may decrease water clarity due to increased biomass growth, abiotic redox reactions, and floc disruption

3.2.1 Micro-aeration caused slight reductions in solid removal efficiencies

In this study, TS and VS removal were evaluated under AD and MA-AD conditions. A significant difference in TS removal efficiency

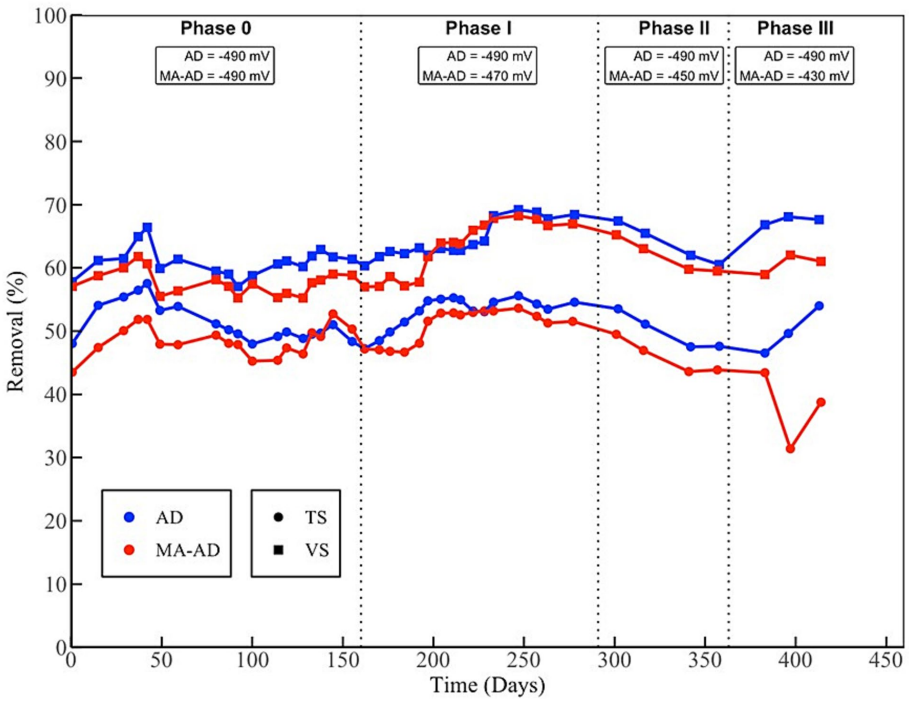


FIGURE 4
Removal efficiency of total solids (TS) and volatile solids (VS) over time in anaerobic digestion (AD) and micro-aerated anaerobic digestion (MA-AD) reactors. The experiment was conducted across four operational phases with distinct oxidation–reduction potential (ORP) setpoints. The AD reactor consistently achieved higher TS and VS removal efficiencies compared to the MA-AD reactor throughout the operational period. TS, total solids; VS, volatile solids; AD, anaerobic digestion; MA-AD, micro-aerated anaerobic digestion.

was observed between the two reactors, with the AD reactor showing a 7.05% greater reduction relative to the MA-AD reactor (Figure 4). On average, TS removal in the AD reactor was 51.78 ± 2.98%, whereas in the MA-AD reactor, it was 48.37 ± 4.36%. VS removal was also statistically different, with the AD reactor achieving 63.08 ± 3.25% and the MA-AD reactor 60.46 ± 4.00%, corresponding to a 4.33% greater reduction in the AD system relative to the MA-AD system (Figure 4). To gain more insight into these dynamics, each operational phase was analyzed separately using independent t-tests. TS removal showed a statistically significant difference in Phase I ($p = 0.0103$), while Phases II ($p = 0.0976$) and III ($p = 0.0521$) were not statistically significant, though the latter approached significance. For VS removal, the differences were not significant in Phase I ($p = 0.4019$) and Phase II ($p = 0.3849$) but became statistically significant in Phase III ($p = 0.0090$). However, in contrast to the findings of Barati et al. (2020), who reported that micro-aeration enhanced solids degradation in anaerobic systems, our results align more closely with those of Piaggio et al. (2023), who observed an increase in solids concentration under micro-aerobic conditions, with no statistically significant improvement in volatile solids.

The reduction in TS removal efficiency in the MA-AD reactor may be attributed to several factors, including an increase in microbial biomass content resulting from heightened cellular growth stimulated by micro-aerobic conditions (Diak et al., 2013) and altered microbial dynamics. These shifts in microbial communities under limited oxygen exposure may have favored biomass accumulation over effective degradation of particulate matter, thereby reducing overall solids removal efficiency. Previous studies have reported that micro-aeration can alter the microbial community structure by enhancing the growth of facultative microorganisms that thrive under low-oxygen conditions (Morais et al., 2024), which could result in different metabolic pathways for the degradation of solids. These pathways, while potentially enhancing the production of organic acids or other metabolites, may not be as efficient at breaking down the bulk solids, including the more recalcitrant fractions of the solids. Specifically, Romero et al. (2021) reported that micro-aeration did not enhance TS removal from AD treatment of sewage sludge from municipal water resource recovery facilities. Furthermore, this suggests that AD was as effective as MA-AD in breaking down the more biodegradable volatile components in the organic matter. Perhaps the advantages of micro-aeration are only realized when high concentrations of less degradable VS are present and not the more easily degradable VS. These results are consistent with findings from Diak et al. (2013), who reported that micro-aeration did not enhance the removal of TS and VS of primary sludge.

From a sustainability perspective, TS and VS are critical water quality metrics as they reflect the particulate and organic load of the treated CPW, which in turn, will affect the usability of downstream freshwater bodies for irrigation and other human activities. Reduced solids removal would result in higher environmental and economic burdens as residual solids contribute to eutrophication, clog irrigation systems, and pose health risks when reused for agriculture or domestic purposes. Thus, while MA-AD offers potential benefits towards the degradation of ecopharmokinetic and biologically recalcitrant compounds, it must be carefully managed to ensure that losses in core treatment performance metrics like TS and VS reduction do not outweigh those benefits.

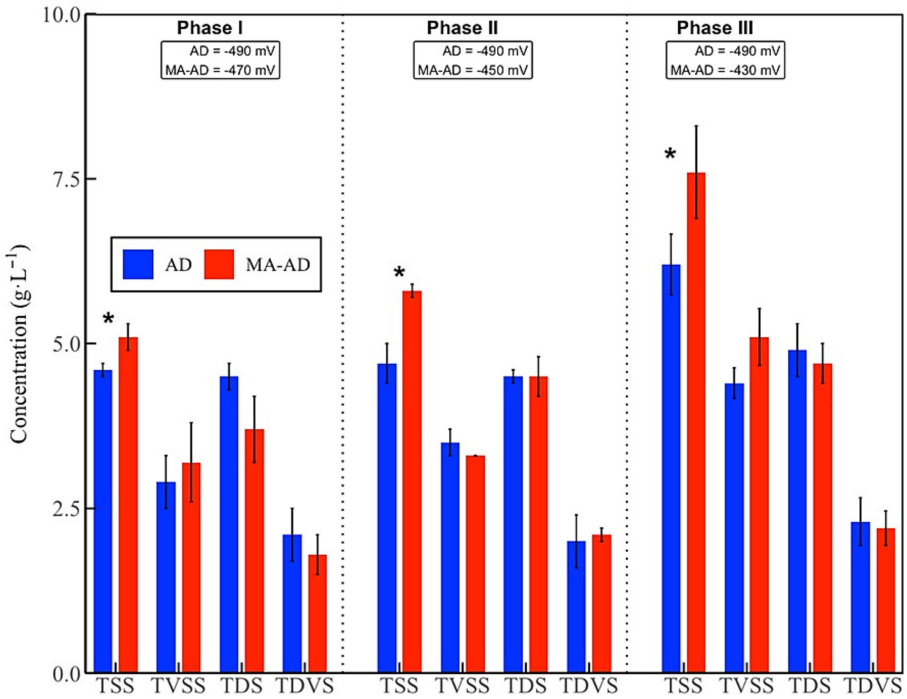


FIGURE 5
Concentrations of total suspended solids (TSS), total volatile suspended solids (TVSS), total dissolved solids (TDS), and total dissolved volatile solids (TDVS) across Phases I–III in anaerobic digestion (AD) and micro-aerated anaerobic digestion (MA-AD) reactors. Among the parameters measured, only TSS levels showed statistically significant differences between the two systems, with MA-AD exhibiting higher TSS concentrations ($p < 0.05$). Error bars represent standard deviation of replicates. AD, anaerobic digestion; MA-AD, micro-aerated anaerobic digestion.

TABLE 3 Effect of micro-aeration on concentrations of dissolved and suspended solids in anaerobic digestion (AD) and micro-aerated anaerobic digestion (MA-AD) reactors across Phases I–III.

Reactor	Phase	TDS ($g \cdot L^{-1}$)	TDVS ($g \cdot L^{-1}$)	TSS ($g \cdot L^{-1}$)	TVSS ($g \cdot L^{-1}$)
AD	I	4.5 ± 0.16	2.1 ± 0.40	4.6 ± 0.10^{a}	2.9 ± 0.40
MA-AD	I	3.7 ± 0.50	1.8 ± 0.30	5.1 ± 0.20^{b}	3.2 ± 0.60
AD	II	4.50 ± 0.50	2.0 ± 0.40	4.7 ± 0.30^{a}	3.5 ± 0.20
MA-AD	II	4.5 ± 0.30	2.1 ± 0.10	5.8 ± 0.10^{b}	3.3 ± 0.00
AD	III	4.9 ± 0.16	2.3 ± 0.36	6.2 ± 0.46^{a}	4.4 ± 0.23
MA-AD	III	4.7 ± 0.20	2.2 ± 0.26	7.6 ± 0.70^{b}	5.1 ± 0.43

Parameters include total dissolved solids (TDS), total dissolved volatile solids (TDVS), total suspended solids (TSS), and total volatile suspended solids (TVSS). Values are reported as mean ± standard deviation ($n = 3$). Within each phase, TSS values with different superscripts (a, b) indicate significant differences ($p < 0.05$, independent samples t-test).

3.2.2 Micro-aeration alters the solids profile of the treated coffee processing water effluent via biological and physico-chemical mechanisms

Effluent samples (50 mL) were collected from the AD and MA-AD reactors at the end of each treatment phase and analyzed in triplicate to evaluate total suspended solids (TSS), total volatile suspended solids (TVSS), total dissolved solids (TDS), and total volatile dissolved solids (TVDS) concentrations. The TSS concentrations in the MA-AD reactor effluent were consistently higher than those observed in the AD reactor across all phases (Figure 5 and Table 3). Specifically, in Phase I, the MA-AD reactor exhibited a TSS concentration of 5.10 ± 0.20 $g \cdot L^{-1}$, while the AD reactor had 4.60 ± 0.10 $g \cdot L^{-1}$. In Phase II, the TSS concentrations were 5.80 ± 0.10 $g \cdot L^{-1}$ for MA-AD and 4.70 ± 0.30 $g \cdot L^{-1}$ for AD. In Phase III, a marked increase in TSS concentrations was observed in the AD and MA-AD reactors. Specifically, the AD reactor was 6.20 ± 0.46 $g \cdot L^{-1}$, and the MA-AD reactor was 7.6 ± 0.70 $g \cdot L^{-1}$. These differences were statistically significant ($p < 0.05$). This consistent upward trend in both treatment conditions suggests that micro-aeration was not the sole factor driving the TSS increase. While the higher final value in the MA-AD reactor may still point to an amplifying effect due to finite oxygen availability, the concurrent rise in the strictly anaerobic control system indicates that other factors were also at play. These may include (1) cumulative biomass accumulation over time, (2) the progressive breakdown of bound or colloidal solids, or (3) reduced sludge settleability due to prolonged reactor operation. The elevated TSS levels in the MA-AD reactor further suggest complex interactions between suspended solids and microbial communities under micro-aerobic conditions (Figure 5). The limited oxygen supply likely promoted the growth of facultative anaerobes, which thrive in low-oxygen environments. Microbial biomass contributes to TVSS, and a numerical increase in TVSS was observed in Phases I and III within the MA-AD reactor relative to the AD reactor, suggesting an increase in biomass growth. This microbial shift may have influenced the flocculation and aggregation of solids, enhancing their retention within the reactor (Liu et al., 2023).

In addition to these biological factors, physicochemical mechanisms may have contributed to the observed increase in TSS under

micro-aerobic conditions. Efendi et al. (2023) reported that increased aeration time and airflow in wastewater treatment systems resulted in a progressive increase in TSS, strongly correlated with DO levels. The study attributed this trend to the reaction between DO and dissolved metal ions, particularly Fe^{2+}, forming insoluble $Fe(OH)_3$ precipitates that increase measured TSS. In this study, Fe^{2+} was added as part of the trace mineral supplement in the feed. Although dosed at relatively low concentrations, the oxidative environment created by micro-aeration likely facilitated the conversion of Fe^{2+} to Fe^{3+}, followed by precipitation as $Fe(OH)_3$, thereby contributing directly to the elevated TSS values observed in the MA-AD reactor. This provides a plausible mechanistic explanation for the significant difference in TSS between the MA-AD and AD reactors in both treatment phases. Moreover, the formation of $Fe(OH)_3$ releases three protons per Fe^{3+} ion, which could partly explain the lower pH levels observed in the MA-AD reactor.

However, Zouari and Al Jabiri (2015) reported contradicting results, noting that micro-aeration improved the digestibility of TSS by enhancing the breakdown of complex organic matter, thus preventing its accumulation in sludge. However, such outcomes may be highly system-dependent, varying with substrate composition, oxygen exposure regimes, and microbial consortia. Furthermore, Jenicek et al. (2011) observed no significant differences in TSS or TVSS concentrations during the micro-aerobic treatment of digested sludge, further suggesting that micro-aeration may not universally enhance solids degradation. Similarly, Uman et al. (2018) observed no significant difference in solids concentrations when applying the Fenton reaction [i.e., H_2O_2 (oxidizer) + Fe^{2+}] to pretreat wastewater biosolids before AD. The elevated TSS in the MA-AD reactor points to the complex microbial and chemical interactions that need to be better controlled to maximize sustainability outcomes. For example, oxygen-enhanced biomass growth and iron precipitation, while not immediately harmful to human health, may lead to greater downstream impacts, including increased costs for sludge handling. It also may render the treated water unsuitable for crop irrigation in rural areas where water reuse is essential for agricultural productivity due to limited access to freshwater resources and commercial fertilizer.

In contrast, TDS, TDVS, and TVSS removal efficiencies showed no statistically significant differences between the AD and MA-AD reactors across treatment phases (Figure 5 and Table 3). These closely aligned values across systems and phases suggest that micro-aeration did not significantly enhance the microbial breakdown of either dissolved or particulate organic solids. These findings support previous reports showing that strictly anaerobic conditions may be more effective for degrading VS due to the dominance of specialized obligate anaerobes that metabolize VFAs and other organics (Appels et al., 2008). In summary, these results demonstrate that careful control of these parameters is essential to realize the full potential of micro-aeration.

3.2.3 Micro-aeration may increase the turbidity of the treated coffee processing water

The impact of micro-aeration on turbidity was evaluated by monitoring changes in nephelometric turbidity units (NTU) across two treatment phases, a key water quality indicator that reflects the presence of suspended colloidal particles, microbial flocs, and inorganic precipitates in the water. In the context of CPW, turbidity is not only a proxy for effluent clarity but also for microbial and physicochemical stability, both of which directly impact downstream reuse potential and environmental impacts, especially in regions where the conventional practice is to discharge CPW directly into water bodies, such as rivers or holding ponds.

In this study, turbidity was monitored across three treatment phases to evaluate the influence of micro-aeration on effluent quality. At the end of each treatment phase, 10 mL of reactor effluent was collected, diluted 1:4 (v/v) with deionized water, and analyzed in triplicate (technical replicates). Under strictly AD conditions, turbidity values were recorded at 2030.22 ± 76.60 NTU in Phase I and 2404.00 ± 162.28 NTU in Phase II (Figure 6). In contrast, under MA-AD conditions, turbidity increased to 2143.33 ± 88.97 NTU in Phase I and significantly more in Phase II, reaching 3146.67 ± 113.89 NTU. In Phase III, this trend became even more pronounced, with turbidity rising to 2525.55 ± 138.45 NTU in the AD reactor and sharply to 6064.22 ± 241.66 NTU in the MA-AD reactor (Figure 6). These findings suggest that the redox conditions and the aeration rate influence turbidity.

In addition, higher redox potential and oxygen availability can influence turbidity outcomes (Zhang et al., 2019). In biological system, this may occur through alterations in microbial metabolism and floc integrity, releasing fine particulate matter into It is also plausible that elevated microbial metabolic activity partially degraded solids into smaller, light-scattering particles, contributing to increased turbidity. Furthermore, under micro-aerobic conditions, physicochemical processes such as oxidation of dissolved sulfur compounds or metals could have contributed to turbidity. For example, Tang et al. (2004) observed that aeration in hydrogen sulfide-rich systems can lead to the formation of insoluble sulfur precipitates, which significantly increase turbidity. The high turbidity recorded in the MA-AD reactor during Phase III (6064.22 ± 241.66 NTU) suggests a threshold beyond which excessive oxygen input or prolonged aeration may induce floc disintegration and increase the release of fine particles. This supports the hypothesis that micro-aeration may trigger similar precipitation pathways in AD, particularly in the presence of sulfur- or metal-containing ions.

In effluents with high organic loads and complex compounds such as CPW, the effect of micro-aeration on turbidity removal may be limited unless specific factors, such as aeration position and oxygen content, are optimized (Zhang et al., 2019). Turbidity directly influences the usability of treated water for irrigation or discharge, which is critical in coffee-producing regions where infrastructure for advanced post-treatment is often lacking. Elevated turbidity diminishes water clarity, adversely affecting aquatic photosynthetic organisms, and increases energy requirements for further filtration and polishing, which may not be economically viable for smallholder operations or cooperatives. Consequently, a trade-off exists that requires the careful weighing of treatment priorities. While co-digestion improves the nutrient balance and increases the biogas yield of AD and MA-AD, it also introduces operational challenges such as increased turbidity and sludge production.

If the objective is to achieve maximum effluent clarity or meet regulatory discharge standards, additional treatment steps such as sedimentation, filtration, or coagulation–flocculation may be necessary to offset the impacts of co-digestion with complex substrates. In contrast, if the primary aim is resource recovery, such as energy generation or enhanced degradation of recalcitrant compounds, higher turbidity levels may be tolerated, provided the effluent is reused in a non-sensitive application or undergoes further low-cost polishing. In conclusion, while micro-aeration is a low-cost, retrofittable enhancement to conventional AD systems, its influence

on turbidity presents a trade-off. While it can positively influence microbial and physicochemical dynamics within AD systems, it also poses a risk of increasing turbidity, particularly under high oxygen dosing conditions. These results highlight the importance of fine-tuning the aeration strategy and substrate composition to balance biological enhancement with the potential destabilization of solids.

3.3 Increased oxygen dosing was associated with lower bioenergy yields

Methane production was consistently higher in the AD reactor across all experimental phases of the study. During the start-up phase (Phase 0), the AD reactor produced approximately 100.0 ± 16.7 mL CH_4·gCOD^{-1}·L^{-1}·d^{-1}, while the MA-AD reactor yielded a comparable 101.9 ± 21.5 mL CH_4·gCOD^{-1}·L^{-1}·d^{-1} (Figure 7), indicating near-equal performance. Notably, the low methane yield observed during the start-up phase in both reactors shows the inherently low biodegradability of the CPW and the co-digestion substrate, WAS. Several studies have reported low methane yields from untreated WAS, typically ranging from 80 to 192 mL CH_4·gVS^{-1}, primarily due to its high concentration of degradation-resistant microbial cells, the low content of non-cellular biodegradable organic matter, and the general need for pretreatment to enhance hydrolysis and microbial accessibility (Feki et al., 2020; Guo et al., 2023; Kampioti and Komilis, 2022; Uman et al., 2018; Wang et al., 2016). Additionally, Widjaja et al. (2017) reported that co-digesting coffee pulp without appropriate pretreatment does not significantly enhance methane yields due to the persistence of lignocellulosic and polyphenolic compounds. In contrast, studies involving AD of readily degradable substrates such as food waste or its co-digestion with energy-rich materials like paper waste, animal manure, or fats, oils, and greases (FOGs) have consistently reported much higher methane yields, typically ranging from 388 to 607 mL CH_4·gVS^{-1} (Kim and Oh, 2011; Marañón et al., 2012; Oduor et al., 2022; Usack and Angenent, 2015; Zhang et al., 2013), showing enhanced biodegradability and the limitations of using recalcitrant substrates such as WAS and CPW without pretreatment.

During the experimental phases, the methane production in the MA-AD reactor dropped to 77.5 ± 13.7 mL CH_4·gCOD^{-1}·L^{-1}·d^{-1} in Phase I, a 23.9% reduction relative to 102.3 ± 12.1 mL CH_4·gCOD^{-1}·L^{-1}·d^{-1} in the AD reactor. This decline became more pronounced in Phase II, where methane production in the MA-AD reactor decreased further to 62.8 ± 6.9 mL CH_4·gCOD^{-1}·L^{-1}·d^{-1}, representing a 43.5% decrease relative to the AD reactor, which generated 111.2 ± 8.3 mL CH_4·gCOD^{-1}·L^{-1}·d^{-1}. In Phase III, the reduced methane production continued, with the AD reactor producing 110.7 ± 19.47 mL CH_4·gCOD^{-1}·L^{-1}·d^{-1}, while methane production in the MA-AD reactor declined further to just 53.4 ± 6.03 mL CH_4·gCOD^{-1}·L^{-1}·d^{-1}, a reduction of nearly 52% compared to the control. These findings are consistent with those of Tsapekos et al. (2021), who observed that while low oxygen doses stimulated methane yield, higher doses exerted a suppressive effect due to the sensitivity of methanogens to oxygen. Micro-aeration can also shift the microbial community in favor of aerobic and facultative bacteria (Nguyen and Khanal, 2018), which directly compete with methanogens for substrates. The volumetric oxygen input required to

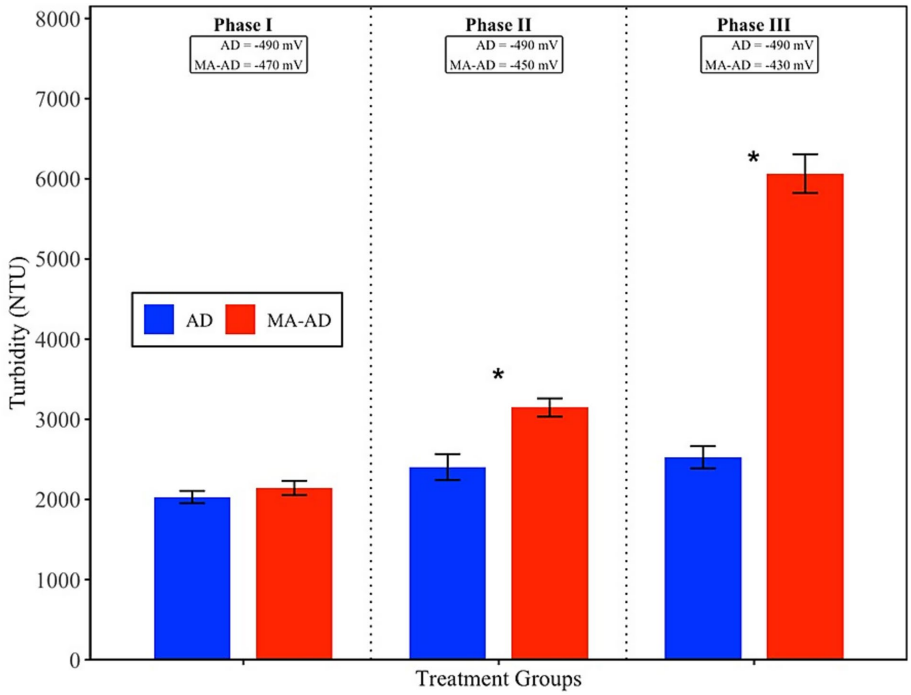


FIGURE 6
Turbidity measurements across Phases I–III in anaerobic digestion (AD) and micro-aerated anaerobic digestion (MA-AD) reactors. While turbidity remained comparable between systems in Phase I, significant increases were observed in the MA-AD reactor during Phases II and III ($p < 0.05$), likely due to increased suspended solids associated with micro-aeration. Error bars represent standard deviation of replicates. AD, anaerobic digestion; MA-AD, micro-aerated anaerobic digestion.

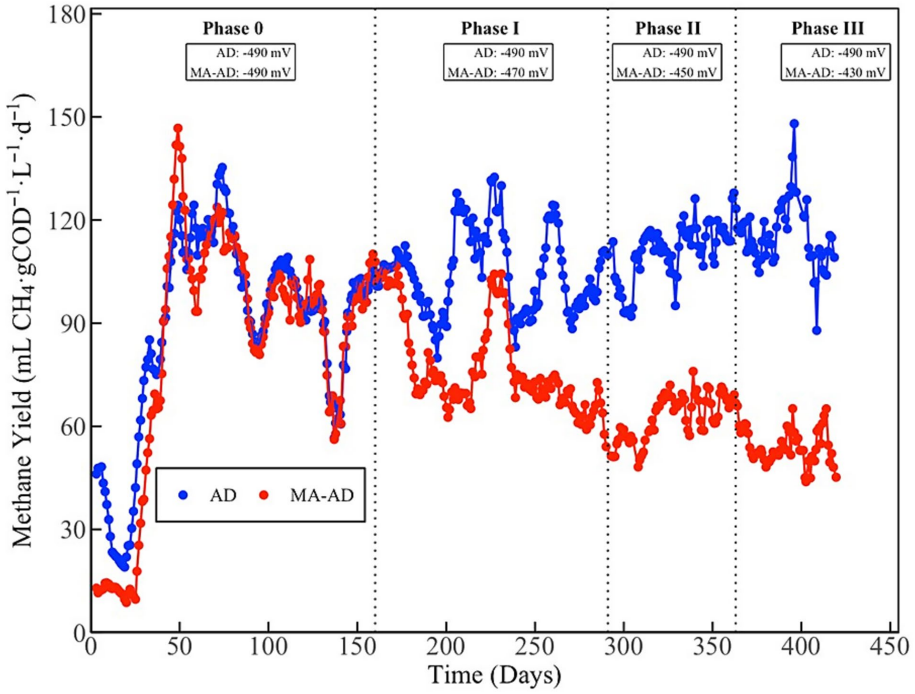


FIGURE 7
Methane yield (mL $CH_4 \cdot gCOD^{-1} \cdot L^{-1} \cdot d^{-1}$) over time in anaerobic digestion (AD) and micro-aerated anaerobic digestion (MA-AD) reactors during four operational phases. While both systems performed similarly during Phase 0, methane production declined in the MA-AD reactor in subsequent phases, indicating a potential inhibitory effect of sustained micro-aeration on methanogenic activity. AD consistently maintained higher methane yield throughout Phases I–III. AD, anaerobic digestion; MA-AD, micro-aerated anaerobic digestion.

hold the redox setpoint increased non-linearly from 28.73 ± 1.10 mL $O_2 \cdot d^{-1} \cdot L_{reactor}^{-1}$ (Phase I) to 100.73 ± 6.90 mL $O_2 \cdot d^{-1} \cdot L_{reactor}^{-1}$ (Phase II) and 228.60 ± 3.92 mL $O_2 \cdot d^{-1} \cdot L_{reactor}^{-1}$ (Phase III) (Figure 8), suggesting the enrichment of oxygen-utilizing bacteria. Interestingly, on Day 404, a sharp decline in oxygen consumption was observed shortly after spiking caffeine, suggesting the caffeine may have temporarily inhibited aerobic respiration by these microbes, thereby reducing oxygen uptake.

Micro-aeration involves modifying the AD microbiome through the controlled dosing of oxygen, which must be done precisely to avoid adverse outcomes (Zhou et al., 2025). Controlled gas dosing is a scalable and widely used practice in wastewater treatment that can be easily integrated into existing AD infrastructure after minor modifications (e.g., the addition of gas blowers and sparger arrays). AD operations that already include basic process control measures, such as controlled feeding (i.e., continuous or semi-continuous), temperature regulation, and constant mixing, are well-positioned for micro-aeration, as they provide a more uniform environment in which to define a precise dosing regimen. Micro-aeration would be less feasible in AD operations involving fewer process control measures, as is often the case in many small-scale AD systems. These simpler AD systems often lack metered feeding systems, instead being fed at irregular intervals, and do not perform active temperature control or constant mixing, resulting in temporal and spatial heterogeneity that would make consistent and uniform gas dosing challenging. In these cases, the AD operator would need to develop a treatment protocol specific to their system. For example, gas dosing could be scheduled based on (1) time of day to track temperature changes, and (2) feeding events to align treatment intensity with substrate loads, thereby achieving more predictable outcomes. Mixing can also be performed during or immediately following the gas dosing cycle using manual agitators (Usack et al., 2014) or electrically-powered impellers or gas recirculation pumps. Lastly, intermittent CPW generation could be managed using equalization or pre-acidification tanks upstream of the AD system, to enable consistent feeding.

In conclusion, while MA-AD can serve as an affordable, low-tech enhancement to conventional AD, its impact on methane production must be carefully evaluated, especially in resource-limited settings where energy recovery is a critical co-benefit. In regions where energy recovery is the goal, maintaining strict anaerobic conditions is likely to maximize biogas output and support on-site clean energy demands. Conversely, when the treatment objective is focused on enhancing hydrolysis, controlling intermediate products such as VFAs, or reducing specific biological recalcitrant compounds to ensure safe environmental discharge or reuse, MA-AD may represent a more robust approach. Here, we found that the lowest oxygen dosing level of ~30 mL $O_2 \cdot d^{-1} \cdot L_{reactor}^{-1}$ provided the best all-around benefits.

3.4 Micro-aeration-assisted degradation of ecopharmokinetic and biologically recalcitrant compounds requires precise redox control and oxygen dosing

This section provides insights into how oxygen exposure and redox dynamics influence the removal efficiency and the degradation behavior of specific ecopharmokinetic and biologically recalcitrant

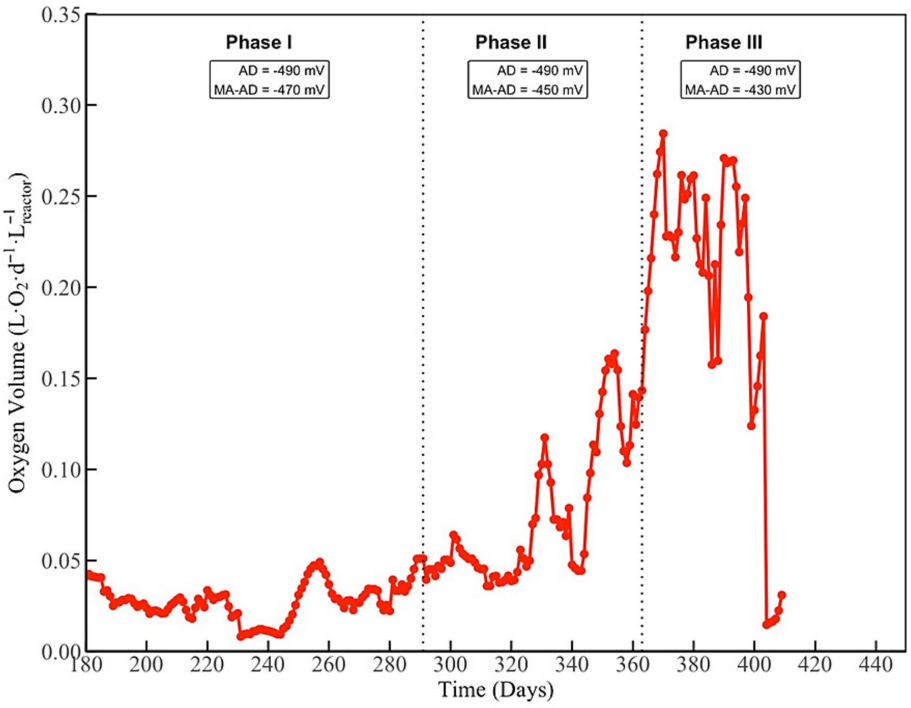


FIGURE 8
Oxygen dosing volume in the micro-aerated anaerobic digestion (MA-AD) reactor during Phases I–III. Oxygen was introduced to maintain specific oxidation–reduction potential (ORP) setpoints, with increasing volumes required in later phases as the ORP target became less negative. The sharp rise in oxygen dosing during Phase III reflects the higher oxygen demand to maintain ORP at −430 mV. No oxygen was added in the conventional anaerobic digestion (AD) reactor.

compounds (caffeine and other chromophoric organics) under strictly AD and MA-AD conditions.

3.4.1 Excess oxygen inhibits the complete breakdown of chromophoric compounds

Ultraviolet (UV) absorbance analysis was employed to quantify the relative concentration of chromophoric organic compounds in the effluents of AD and MA-AD reactors, using raw CPW as a calibration reference. An initial full-spectrum scan of the undiluted CPW revealed a maximum absorbance at 270 nm, a signature wavelength for polyphenols, aromatic rings, and other conjugated chromophores commonly present in agro-industrial waste streams. Serial dilutions of CPW were prepared to construct a calibration curve, from which the relative concentrations of UV-absorbing compounds in reactor effluents were back-calculated.

Effluent samples collected at the end of Phases I–III showed varied levels of residual chromophoric compounds. In Phase I, the relative concentration was 30.09 ± 1.13% in the AD reactor and 19.53 ± 0.82% in the MA-AD reactor, indicating slightly better removal under micro-aerobic conditions (Figure 9). However, in Phase II, the concentration in the MA-AD reactor spiked to 92.46 ± 3.06%, compared to 39.67 ± 1.94% in the AD reactor, suggesting a substantial decline in the degradation of chromophoric compounds under increased oxygen input. In Phase III, the MA-AD reactor again showed a higher residual concentration (70.11 ± 2.32%) compared to the AD reactor (42.51 ± 1.90%), showing that oxygen dosing at higher levels may impair the removal of UV-absorbing species. These trends are consistent with the findings from caffeine degradation trends discussed in Section 3.4.2, where the MA-AD reactor exhibited enhanced early-stage degradation kinetics but failed to achieve complete removal in later stages. This parallel behavior suggests a shared mechanistic limitation, potentially involving disruption of obligate anaerobic pathways or an oxygen-induced metabolic shift that favored partial oxidation over complete degradation.

Optimizing MA-AD systems for the effective removal of UV-absorbing, aromatic-rich organics, such as those found in CPW, requires a careful balance of oxygen input and microbial community management. Low-level micro-aeration can stimulate the oxidative breakdown of complex compounds, while excessive oxygen exposure disrupts anaerobic pathways critical for complete degradation. Over-aeration reduces treatment efficiency and increases energy consumption, leading to additional operational costs in industrial applications and straining energy resources in decentralized or rural systems. It also diminishes effluent quality, potentially hindering its safe reuse in agriculture or environmentally sensitive contexts. The adoption of low-energy, sensor-controlled aeration strategies, combined with precise ORP regulation, can facilitate stable, resource-efficient, and environmentally sustainable wastewater treatment, especially in low-infrastructure settings where centralized treatment is unavailable.

3.4.2 Micro-aeration accelerates caffeine breakdown kinetics at low dosing levels

For Phase I, the MA-AD reactor showed faster caffeine degradation compared to the AD reactor. Within 28 h, caffeine concentrations decreased by >85% in the MA-AD reactor, while the AD reactor achieved approximately 75% removal (Figure 10). This

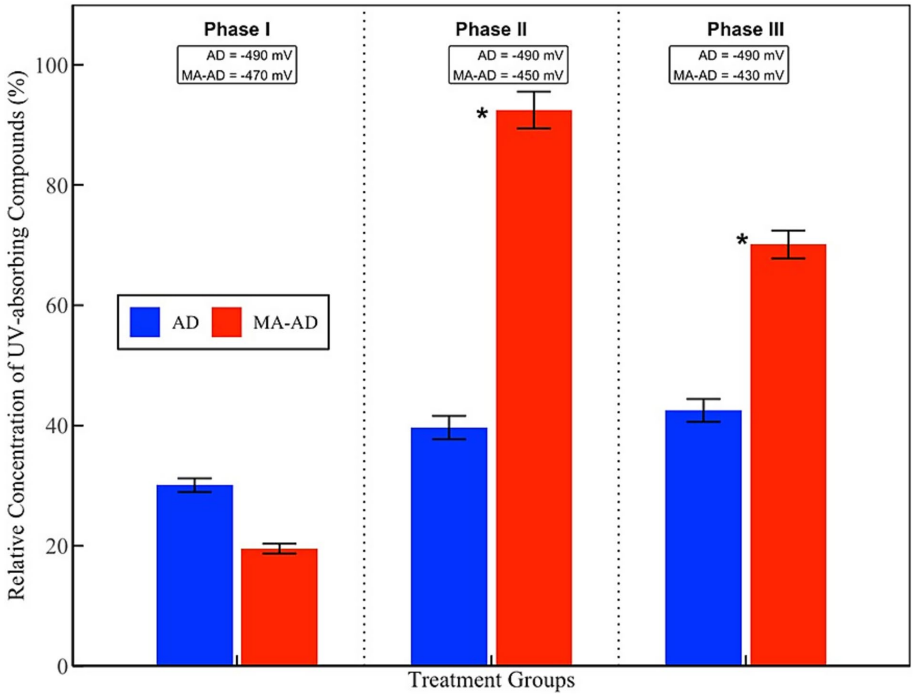


FIGURE 9
Relative concentration of UV-absorbing compounds in anaerobic digestion (AD) and micro-aerated anaerobic digestion (MA-AD) reactors across treatment phases. While both systems showed comparable or slightly lower concentrations during Phase I, significantly elevated levels were observed in the MA-AD reactor during Phases II and III ($p < 0.05$), suggesting accumulation of refractory or aromatic intermediates under micro-aerobic conditions. Error bars indicate standard deviations of technical replicates. AD, anaerobic digestion; MA-AD, micro-aerated anaerobic digestion.

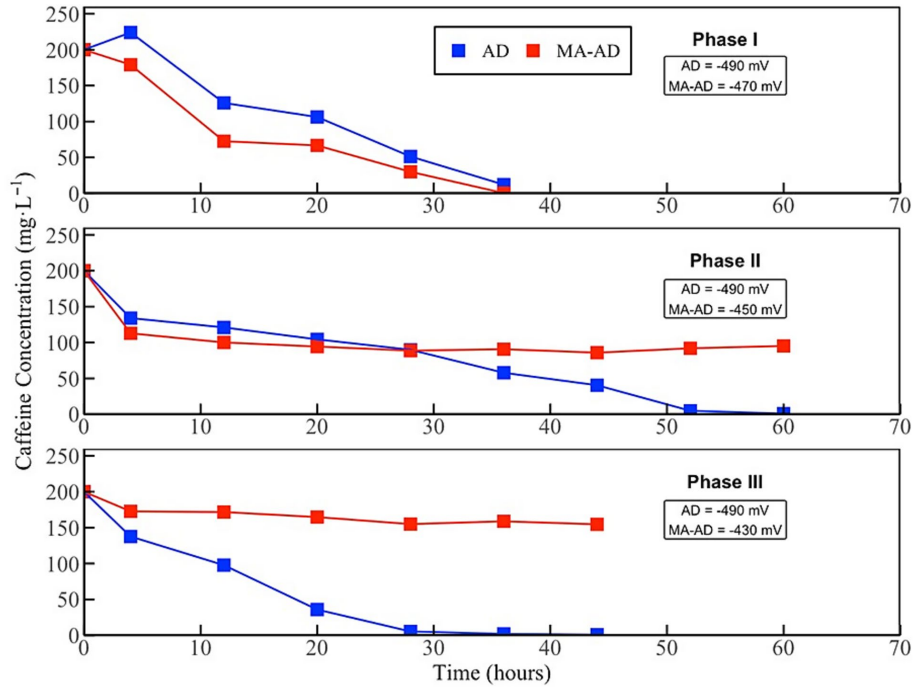


FIGURE 10
Caffeine degradation curve in anaerobic digestion (AD) and micro-aerated anaerobic digestion (MA-AD) reactors across Phases I–III. In Phase I, the MA-AD system demonstrated a faster initial degradation rate compared to AD. However, in Phases II and III, caffeine removal in MA-AD plateaued after the initial faster reduction, while the AD reactor continued to achieve greater overall degradation. These observations indicate that micro-aeration may enhance short-term removal but hinder sustained biodegradation under prolonged operation. AD, anaerobic digestion; MA-AD, micro-aerated anaerobic digestion.

initial enhancement is likely due to the stimulation of facultative microbial activity, which has been shown to accelerate the breakdown of aromatic compounds under oxygen-limited conditions (Aydin et al., 2025). Facultative microorganisms capable of co-metabolizing caffeine through oxidative demethylation may have been enriched under these conditions, contributing to the observed advantage. However, by 36 h, both reactors had reached near-complete degradation (~90%), indicating convergence of long-term removal efficiency regardless of oxygen input. A similar trend was reported by Chen et al. (2018), who demonstrated that significant caffeine reductions can be achieved under strictly anaerobic conditions using an anaerobic membrane reactor, reporting a long-term removal efficiency of 87.5 ± 5.3%. In their study, caffeine degradation followed a two-step methanogenic pathway: the initial hydrolysis of caffeine into intermediates, followed by conversion of those intermediates into VFAs, and eventually CH_4 and CO_2. Notably, the transformation of hydrolysis products into VFAs was identified as the rate-limiting step of the process. Hydrolysis may have been rate limiting in the present study, given the lower SCOD concentration in the MA-AD reactor, and assuming a faster rate of product metabolism.

In Phase II, both reactors were spiked again after reaching baseline levels. Once more, the MA-AD reactor demonstrated a faster initial degradation rate than the strictly AD reactor. By 28 h, the caffeine concentration in the MA-AD reactor had decreased to 94.4 ± 0.2 mg·L^{-1}, compared to 104.4 ± 0.3 mg·L^{-1} in the AD reactor (Figure 10). This early-phase degradation advantage supports earlier observations that micro-aeration may enhance the initial transformation of caffeine, likely due to the stimulation of facultative microbial populations capable of oxidative co-metabolism. However, this initial performance advantage was not sustained. In Phase II, beyond 36 h, the trend reversed. The AD reactor showed continued degradation, reaching near-complete removal (~1.05 mg·L^{-1}) by 60 h, while caffeine levels in the MA-AD reactor stagnated between 85–95 mg·L^{-1} from 36 h onward (Figure 10). Furthermore, throughout this stage, the volume of oxygen dosed into the MA-AD reactor increased markedly compared to earlier phases (Figure 8). This increase may reflect shifts in microbial oxygen demand or reduced microbial responsiveness to redox levels. Such excessive oxygenation could inhibit obligate anaerobes, impairing the conversion of caffeine intermediates into VFAs and methane (Lu and Imlay, 2021).

In Phase III, both reactors were again spiked with 200 mg·L^{-1} of caffeine. Caffeine removal in the MA-AD reactor remained incomplete, with concentrations only declining from 200 mg·L^{-1} to 172.68 ± 3.38 mg·L^{-1} after 4 h and finally plateauing at 132.36 ± 0.08 mg·L^{-1} after 60 h, like the trend observed in Phase II. Meanwhile, the AD reactor showed consistent degradation, reaching near-complete removal (0.95 ± 0.38 mg·L^{-1}) by 44 h. This may imply that the microbial population in the MA-AD reactor had shifted away from specific microbial species capable of degrading caffeine or that an accumulation of intermediate metabolites from previous stages interfered with degradation. Alternatively, increased oxygen pressure may have limited the activity of facultative anaerobes without fully re-establishing anaerobic syntrophy.

Although aerobic microorganisms generally exhibit greater metabolic efficiency and higher energy yields than their anaerobic counterparts, their ability to degrade caffeine in the MA-AD reactor may have been constrained by several factors. The microbial community present may not have been fully adapted to aerobic caffeine degradation, requiring specific enzymes or co-metabolic conditions that were not prevalent in the MA-AD reactor. Furthermore, oxygen toxicity may have inhibited certain microbial groups that play a crucial role in caffeine breakdown, disrupting enzymatic pathways and leading to metabolic inefficiencies. Additionally, aerobic pathways may lead to the accumulation of intermediate metabolites that were not readily degraded under the given conditions, potentially inhibiting further microbial activity. These findings indicate the importance of oxygen dosing precision and microbial community acclimation in maintaining the effectiveness of micro-aerated systems for recalcitrant compound degradation.

From a sustainability perspective, this has critical implications: caffeine is a persistent, bioactive compound that can pass through conventional wastewater treatment systems and pose risks to aquatic ecosystems and downstream human populations relying on surface or groundwater for drinking or irrigation. The inconsistent degradation observed in the MA-AD reactor under higher oxygen loadings suggests that poorly calibrated micro-aeration may reduce the long-term reliability of contaminant removal. In an industrial setting, this introduces a risk of regulatory non-compliance and higher treatment costs if additional polishing steps are required. In decentralized or rural systems, where effluent reuse is often essential for irrigation or fertigation, residual caffeine could adversely impact soils or crops, undermining food security and farmer incomes. Therefore, optimizing caffeine degradation in MA-AD systems is not only a question of process efficiency but a sustainability imperative that intersects with environmental protection, economic viability, and public health. Achieving this requires tuning oxygen input to avoid inhibiting anaerobic syntrophy, supporting microbial acclimation, and ensuring robust performance across operational scales.

4 Conclusion and future perspective

This study comprehensively evaluates MA-AD in treating CPW, focusing on its effects on reactor stability, solids removal, turbidity, methane yield, and the degradation of problematic compounds such as caffeine. The results demonstrate that while micro-aeration can enhance early-stage microbial activity, hydrolysis, and degradation rates of certain recalcitrant compounds, its long-term success depends on carefully balancing operational parameters. When oxygen is applied in low, controlled quantities, it can stimulate facultative microbial activity, promote partial oxidation of complex substrates, and accelerate the initial breakdown of polyphenolic compounds. However, excessive oxygen dosing can disrupt anaerobic microbial consortia, inhibit key methanogenic pathways, and diminish methane yields. This highlights the importance of finely tuned ORP regulation and real-time monitoring to prevent oxygen overload. Furthermore, the trace mineral composition of the substrate plays a pivotal role in reactor behavior. In this study, the presence of Fe^{2+} and other micronutrients may have catalyzed physicochemical reactions such as floc formation and particulate precipitation, influencing turbidity and solids retention. These interactions underscore the interconnectedness between substrate chemistry and reactor performance, especially under shifting redox levels.

The biologically inhibitory effects of CPW were also evident in this study—relatively low concentrations of CPW exerted significant microbial stress. Moreover, the observed shifts in pH and microbial

responses further reinforce the need for cautious substrate formulation and reactor operation. Pretreatment strategies, such as alkaline hydrolysis, enzymatic conditioning, or strategic co-digestion, may offer viable approaches to enhance substrate bioavailability and mitigate inhibition. Ultimately, the success of MA-AD hinges on the integration of operational control (oxygen and ORP), feedstock chemistry (trace elements and inhibitors), and process design (OLR, mixing, and pretreatment). Future research should aim to define optimal oxygen loading thresholds, characterize the microbial community dynamics, and deploy advanced analytical tools (e.g., LC-MS, metabolomics) to track degradation intermediates and byproducts. In conclusion, when precisely managed, MA-AD holds significant potential as an alternative, low-cost treatment modality for coffee processing water; however, further research should be performed to ensure more predictable and robust outcomes before putting the technology into practice in these coffee-producing regions.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

KT: Conceptualization, Investigation, Writing – original draft. SO: Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing. WK: Resources, Writing – review & editing. RP: Resources, Writing – review & editing. JS: Resources, Writing – review & editing. JU: Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing.

Funding

The author(s) declare that financial support was received for the research and/or publication of this article. This work was supported in part by the Agricultural Experiment Station at the University of Georgia, the Office of Global Engagement at the University of Georgia through the Global Research Collaboration Grant program, and the Institute for Integrative Precision Agriculture at the University of Georgia through the Seed Grant program.

Acknowledgments

KT would like to acknowledge Elizabeth Ziabtchenko, Ibrahim Bello, Shreya Riswadkar, and Nithya Sree Kotha for their help running the reactors.

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The authors declare that no Gen AI was used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

Alemayehu, Y. A., Asfaw, S. L., and Tirfie, T. A. (2020). Management options for coffee processing wastewater. A review. *J. Mater. Cycles Waste Manag.* 22, 454–469. doi: 10.1007/s10163-019-00953-y

American Public Health Association (2023). Standard methods for the examination of water and wastewater. 24th Edn. Washington, DC: APHA-AWWA-WEF.

Angenent, L. T., Usack, J. G., Sun, T., Fink, C., Molitor, B., Labatut, R., et al. (2022). "Upgrading anaerobic digestion within the energy economy–the methane platform" in Resource recovery from water: principles and application (London: IWA Publishing), 141–158.

Appels, L., Baeyens, J., Degrève, J., and Dewil, R. (2008). Principles and potential of the anaerobic digestion of waste-activated sludge. *Prog. Energy Combust. Sci.* 34, 755–781. doi: 10.1016/j.pecs.2008.06.002

Aydin, D. C., Aldas-Vargas, A., Grotenhuis, T., and Rijnaarts, H. (2025). Microaerobic biodegradation of aromatic hydrocarbon mixtures: strategies for efficient nitrate and oxygen dosage. *Appl. Microbiol. Biotechnol.* 109:9. doi: 10.1007/s00253-024-13388-9

Barati, R., Rezaee, A., Farzadkia, M., Gholami, M., and Kermani, M. (2020). Effect of micro-aerobic process on improvement of anaerobic digestion sewage sludge treatment: flow cytometry and ATP assessment. *RSC Adv.* 10, 35718–35728. doi: 10.1039/D0RA05540A

Canul Bacab, F., España Gamboa, E., Ruiz Espinoza, J. E., Leal-Bautista, R. M., Tapia Tussell, R., Domínguez Maldonado, J., et al. (2020). Two phase anaerobic digestion system of municipal solid waste by utilizing microaeration and granular activated carbon. *Energies* 13:933. doi: 10.3390/en13040933

Chen, R., Jiang, H., and Li, Y.-Y. (2018). Caffeine degradation by methanogenesis: efficiency in anaerobic membrane bioreactor and analysis of kinetic behavior. *Chem. Eng. J.* 334, 444–452. doi: 10.1016/j.cej.2017.10.052

Dadi, D., Mengistie, E., Terefe, G., Getahun, T., Haddis, A., Birke, W., et al. (2018). Assessment of the effluent quality of wet coffee processing wastewater and its influence on downstream water quality. *Ecohydrol. Hydrobiol.* 18, 201–211. doi: 10.1016/j.ecohyd.2017.10.007

Diak, J., Örmeci, B., and Kennedy, K. J. (2013). Effect of micro-aeration on anaerobic digestion of primary sludge under septic tank conditions. *Bioprocess Biosyst. Eng.* 36, 417–424. doi: 10.1007/s00449-012-0798-x

Díaz, I., Donoso-Bravo, A., and Fdz-Polanco, M. (2011). Effect of microaerobic conditions on the degradation kinetics of cellulose. *Bioresour. Technol.* 102, 10139–10142. doi: 10.1016/j.biortech.2011.07.096

Ding, K., Wu, B., Wang, Y., Xu, L., Liu, M., Xiang, J., et al. (2024). Study on synergistic effect of carrier combined with micro-aeration on anaerobic digestion of food waste. *Chem. Eng. J.* 498:155731. doi: 10.1016/j.cej.2024.155731

Du, N., Li, M., Zhang, Q., Ulsido, M. D., Xu, R., and Huang, W. (2020). Study on the biogas potential of anaerobic digestion of coffee husks wastes in Ethiopia. *Waste Manag. Res.* 39, 291–301. doi: 10.1177/0734242X20939619

Duarte, M. S., Fernandes, R. J., Sousa, J., Pereira, C., Mesquita, D. P., and Alves, M. M. (2024). Influence of micro-aeration in the production of volatile fatty acids (VFA) from wastewaters with high salinity. *Fuel* 375:132566. doi: 10.1016/j.fuel.2024.132566

Efendi, D. A. M. N., Ramandani, A. A., Cendekia, D., and Hanifah, W. (2023). Industrial wastewater treatment using venture injector type micro-bubble aeration as a reduction of dissolved iron (Fe^{2+}) levels. *J. Nat. Sci. Math. Res.* 9, 91–101. doi: 10.21580/jnsmr.2023.9.2.17594

Feki, E., Battimelli, A., Sayadi, S., Dhouib, A., and Khoufi, S. (2020). High-rate anaerobic digestion of waste activated sludge by integration of electro-Fenton process. *Molecules* 25:626. doi: 10.3390/molecules25030626

Fernández-Domínguez, D., Yekta, S. S., Hedenström, M., Patureau, D., and Jimenez, J. (2023). Deciphering the contribution of microbial biomass to the properties of dissolved and particulate organic matter in anaerobic digestates. *Sci. Total Environ.* 877:162882. doi: 10.1016/j.scitotenv.2023.162882

Figueroa Campos, G. A. (2022). "Wet-coffee processing production wastes: quality, potentials, and valorization opportunities" in Doctoral thesis (Potsdam: Universität Potsdam).

Fu, S., Lian, S., Angelidaki, I., and Guo, R. (2023). Micro-aeration: an attractive strategy to facilitate anaerobic digestion. *Trends Biotechnol.* 41, 714–726. doi: 10.1016/j.tibtech.2022.09.008

Gaballah, E. S., Gao, L., Shalaby, E. A., Yang, B., Sobhi, M., Ali, M. M., et al. (2025). Performance and mechanism of a novel hydrolytic bacteria pretreatment to boost waste activated sludge disintegration and volatile fatty acids production during acidogenic fermentation. *J. Environ. Manag.* 376:124399. doi: 10.1016/j.jenvman.2025.124399

Gomes de Barros, V., Rodrigues, C. S. D., Botello-Suárez, W. A., Duda, R. M., Alves de Oliveira, R., da Silva, E. S., et al. (2020). Treatment of biodigested coffee processing wastewater using Fenton's oxidation and coagulation/flocculation. *Environ. Pollut.* 259:113796. doi: 10.1016/j.envpol.2019.113796

Guo, H., Tian, L., Wang, Y., Zheng, K., Hou, J., Zhao, Y., et al. (2023). Enhanced anaerobic digestion of waste activated sludge with periodate-based pretreatment. *Environ. Sci. Ecotechnology* 13:100208. doi: 10.1016/j.ese.2022.100208

Haddis, A., and Devi, R. (2008). Effect of effluent generated from coffee processing plant on the water bodies and human health in its vicinity. *J. Hazard. Mater.* 152, 259–262. doi: 10.1016/j.jhazmat.2007.06.094

Harirchi, S., Wainaina, S., Sar, T., Nojoumi, S. A., Parchami, M., Parchami, M., et al. (2022). Microbiological insights into anaerobic digestion for biogas, hydrogen or volatile fatty acids (VFAs): a review. *Bioengineered* 13, 6521–6557. doi: 10.1080/21655979.2022.2035986

Ijanu, E. M., Kamaruddin, M. A., and Norashiddin, F. A. (2020). Coffee processing wastewater treatment: a critical review on current treatment technologies with a proposed alternative. *Appl Water Sci* 10:11. doi: 10.1007/s13201-019-1091-9

Jenicek, P., Celis, C., Koubova, J., and Ruzickova, I. (2011). Change of the digested sludge quality at microaerobic digestion. *J. Residuals Sci. Technol.* 8, 39–44. Available online at: https://www.scopus.com/pages/publications/79958780627?inward

Kampioti, A., and Komilis, D. (2022). Anaerobic co-digestion of coffee waste with other organic substrates: a mixture experimental design. *Chemosphere* 297:134124. doi: 10.1016/j.chemosphere.2022.134124

Kim, D.-H., and Oh, S.-E. (2011). Continuous high-solids anaerobic co-digestion of organic solid wastes under mesophilic conditions. *Waste Manag.* 31, 1943–1948. doi: 10.1016/j.wasman.2011.05.007

Lim, J. W., and Wang, J.-Y. (2013). Enhanced hydrolysis and methane yield by applying microaeration pretreatment to the anaerobic co-digestion of brown water and food waste. *Waste Manag.* 33, 813–819. doi: 10.1016/j.wasman.2012.11.013

Liu, M., Wang, J., and Peng, Z. (2023). Effects of micro-bubble aeration on the pollutant removal and energy-efficient process in a floc–granule sludge coexistence system. *Water Sci. Technol.* 88, 3044–3055. doi: 10.2166/wst.2023.376

Lu, Z., and Imlay, J. A. (2021). When anaerobes encounter oxygen: mechanisms of oxygen toxicity, tolerance and defence. *Nat. Rev. Microbiol.* 19, 774–785. doi: 10.1038/s41579-021-00583-y

Magdalena, J. A., Angenent, L. T., and Usack, J. G. (2022). The measurement, application, and effect of oxygen in microbial fermentations: focusing on methane and carboxylate production. *Fermentation* 8:138. doi: 10.3390/fermentation8040138

Marañón, E., Castrillón, L., Quiroga, G., Fernández-Nava, Y., Gómez, L., and García, M. (2012). Co-digestion of cattle manure with food waste and sludge to increase biogas production. *Waste Manag.* 32, 1821–1825. doi: 10.1016/j.wasman.2012.05.033

Mazzafera, P. (2002). Degradation of caffeine by microorganisms and potential use of decaffeinated coffee husk and pulp in animal feeding. *Sci. Agric.* 59, 815–821. doi: 10.1590/S0103-90162002000400030

Morais, B. P., Magalhães, C. P., Martins, G., Pereira, M. A., and Cavaleiro, A. J. (2024). Effect of micro-aeration on syntrophic and methanogenic activity in anaerobic sludge. *Appl. Microbiol. Biotechnol.* 108:192. doi: 10.1007/s00253-023-12969-4

Nanjundaiah, S., Mutturi, S., and Bhatt, P. (2017). Modeling of caffeine degradation kinetics during cultivation of *Fusarium solani* using sucrose as co-substrate. *Biochem. Eng. J.* 125, 73–80. doi: 10.1016/j.bej.2017.05.018

Nguyen, D., and Khanal, S. K. (2018). A little breath of fresh air into an anaerobic system: how microaeration facilitates anaerobic digestion process. *Biotechnol. Adv.* 36, 1971–1983. doi: 10.1016/j.biotechadv.2018.08.007

Oduor, W. W., Wandera, S. M., Murunga, S. I., and Raude, J. M. (2022). Enhancement of anaerobic digestion by co-digesting food waste and water hyacinth in improving treatment of organic waste and bio-methane recovery. *Heliyon* 8:e10580. doi: 10.1016/j.heliyon.2022.e10580

Piaggio, A. L., Sasidhar, K., Khande, P., Balakrishnan, M., van Lier, J. B., de Kreuk, M. K., et al. (2023). Effects of low oxygen dosages on an anaerobic membrane bioreactor, simulating the oxygen load in an anaerobic digester-dissolved air flotation (AD-DAF) system. *ACS ES T Water* 3, 4133–4142. doi: 10.1021/acsestwater.3c00544

Qiao, W., Mohammad, S., Takayanagi, K., and Li, Y. (2015). Thermophilic anaerobic co-digestion of coffee grounds and excess sludge: long term process stability and energy production. *RSC Adv.* 5, 26452–26460. doi: 10.1039/C4RA15581E

Ratchawat, T., Panyatona, S., Nopchinwong, P., Chidthaisong, A., and Chiarakorn, S. (2020). Carbon and water footprint of Robusta coffee through its production chains in Thailand. *Environ. Dev. Sustain.* 22, 2415–2429. doi: 10.1007/s10668-018-0299-4

Rattan, S., Parande, A., Nagaraju, V., and Ghiwari, G. K. (2015). A comprehensive review on utilization of wastewater from coffee processing. *Environ. Sci. Pollut. Res.* 22, 6461–6472. doi: 10.1007/s11356-015-4079-5

Rodriguez-Narvaez, O. M., Peralta-Hernandez, J. M., Goonetilleke, A., and Bandala, E. R. (2017). Treatment technologies for emerging contaminants in water: a review. *Chem. Eng. J.* 323, 361–380. doi: 10.1016/j.cej.2017.04.106

Romero, A., Cesca, J., Van Rys, D., Johnson, B., and Kraakman, B. (2021). Process understanding of full-scale micro-aeration to improve biogas quality and anaerobic digestion. *Residuals and Biosolids Conference 2021*,

Selvamurugan, M., Doraisamy, P., and Maheswari, M. (2010). An integrated treatment system for coffee processing wastewater using anaerobic and aerobic process. *Ecol. Eng.* 36, 1686–1690. doi: 10.1016/j.ecoleng.2010.07.013

Tang, Y., Shigematsu, T., Ikbal, Morimura, S., and Kida, K. (2004). The effects of micro-aeration on the phylogenetic diversity of microorganisms in a thermophilic anaerobic municipal solid-waste digester. *Water Res.* 38, 2537–2550. doi: 10.1016/j.watres.2004.03.012

Tsapekos, P., Alvarado-Morales, M., Kougias, P. G., Treu, L., and Angelidaki, I. (2021). Enhancing anaerobic digestion of agricultural residues by microaerobic conditions. *Biomass Convers. Biorefinery* 11, 2325–2333. doi: 10.1007/s13399-019-00430-4

Uman, A. E., Usack, J. G., Lozano, J. L., and Angenent, L. T. (2018). Controlled experiment contradicts the apparent benefits of the Fenton reaction during anaerobic digestion at a municipal wastewater treatment plant. *Water Sci. Technol.* 78, 1861–1870. doi: 10.2166/wst.2018.362

United States Department of Agriculture-Foreign Agricultural Service (2024). Coffee: world markets and trade. Available online at: https://www.fas.usda.gov/data/coffee-world-markets-and-trade-12182024. (Accessed May 5, 2025)

Usack, J., and Angenent, L. (2015). Comparing the inhibitory thresholds of dairy manure co-digesters after prolonged acclimation periods: part 1–performance and operating limits. *Water Res.* 87, 446–457. doi: 10.1016/j.watres.2015.05.055

Usack, J. G., Spirito, C. M., and Angenent, L. T. (2012). Continuously-stirred anaerobic digester to convert organic wastes into biogas: system setup and basic operation. *J. Vis. Exp.* 65:e3978. doi: 10.3791/3978

Usack, J. G., Wiratni, W., and Angenent, L. T. (2014). Improved design of anaerobic digesters for household biogas production in Indonesia: one cow, one digester, and one hour of cooking per day. *Sci. World J.* 2014:318054. doi: 10.1155/2014/318054

Utrilla-Catalan, R., Rodríguez-Rivero, R., Narvaez, V., Díaz-Barcos, V., Blanco, M., and Galeano, J. (2022). Growing inequality in the coffee global value chain: a complex network assessment. *Sustainability* 14:672. doi: 10.3390/su14020672

Vandeponseele, A., Draye, M., Piot, C., and Chatel, G. (2021). Study of influential parameters of the caffeine extraction from spent coffee grounds: from brewing coffee method to the waste treatment conditions. *Clean Technol.* 3, 335–350. doi: 10.3390/cleantechnol3020019

Wang, Q., Sun, J., Zhang, C., Xie, G.-J., Zhou, X., Qian, J., et al. (2016). Polyhydroxyalkanoates in waste activated sludge enhances anaerobic methane production through improving biochemical methane potential instead of hydrolysis rate. *Sci. Rep.* 6:19713. doi: 10.1038/srep19713

Widjaja, T., Iswanto, T., Altway, A., Shovitri, M., and Juliastuti, S. R. (2017). Methane production from coffee pulp by microorganism of rumen fluid and cow dung in co-digestion. *Chem. Eng. Trans.* 56, 1465–1470. doi: 10.3303/CET1756245

Xu, H., Li, Y., Hua, D., Zhao, Y., Chen, L., Zhou, L., et al. (2021). Effect of microaerobic microbial pretreatment on anaerobic digestion of a lignocellulosic substrate under controlled pH conditions. *Bioresour. Technol.* 328:124852. doi: 10.1016/j.biortech.2021.124852

Zhang, C., Xiao, G., Peng, L., Su, H., and Tan, T. (2013). The anaerobic co-digestion of food waste and cattle manure. *Bioresour. Technol.* 129, 170–176. doi: 10.1016/j.biortech.2012.10.138

Zhang, W., Zhang, R., Yang, Y., Huang, T., and Wen, G. (2019). Removal of high concentrations of ammonium from groundwater in a pilot-scale system through aeration at the bottom layer of a chemical catalytic oxidation filter. *Int. J. Environ. Res. Public Health* 16:3989. doi: 10.3390/ijerph16203989

Zhou, M., Bello, I. O., Magdalena, J. A., and Usack, J. G. (2025). Development and biological validation of a flat-sheet membrane bioreactor for ultra-low gas flux bioprocesses. *Chem. Eng. J.* 520:166429. doi: 10.1016/j.cej.2025.166429

Zhou, M., Taiwo, K., Wang, H., Ntihuga, J.-N., Angenent, L. T., and Usack, J. G. (2024). Anaerobic digestion of process water from hydrothermal treatment processes: a review of inhibitors and detoxification approaches. *Bioresour. Bioprocess.* 11:47. doi: 10.1186/s40643-024-00756-6

Zouari, N., and Al Jabiri, H. (2015). Improvement by micro-aeration of anaerobic digestion of slaughterhouse wastewater at 38 °C. *Int. J. Innov. Sci. Eng. Technol.* 4, 807–816. doi: 10.15680/IJIRSET.2015.0403002